
Introduction to Healthcare in the Context of Ageing Societies: A Structured Essay

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Healthcare is changing at a rapid pace: technology is evolving, and new treatments are being implemented at a breakneck pace on a global scale. However, healthcare still remains limited in many aspects. One critical aspect is health and wellbeing for the elderly population. While technology enabling effective care for vulnerable populations has increased, many needs remain unmet. Among them are access to healthcare, effective nutrition and food technology, availability of tools for medication management and rehabilitation, support for mental health, as well as community-driven care. This booklet aims to explore these topics briefly and provide a foundation to solve issues at large.

NOTE: This text was written as a part of the SCORE Research Camp x Hackathon 2025/26. It is intended as only a brief overview of the topic for hackathon participants. Please perform your own research to support ideas being pitched. Furthermore, please do not limit innovations to the topics discussed within this document.

1 Ageing Societies

In the 21st century, both developed and developing countries are rapidly aging. The percentage of citizens aged over 60 is rising in almost all developed countries worldwide. This demographic trend poses a challenge for a wide multitude of sectors, including finance, entertainment, and healthcare. Thailand is no exception to the aging society trend, as it is rapidly transitioning into a super-aged society, with 28% of its population

expected to be aged 60 or older by 2030. This shift is transforming every aspect of Thai society—from healthcare and financial systems to economic productivity and family structures.

This transition does not occur in a vacuum: each affected aspect interacts with each other and often results in great additive effects. One commonly cited example in economics is the indirect effect of an ageing population on the labour supply, which is best explained by the requirements imposed resulting in a restructuring of roles within the family unit (where working-aged adults are moved away from the labour force towards caregiving). While not directly relevant to the present context of this hackathon, such effects do not limit their influence to merely the economic sector and labour force. Healthcare as a sector is also heavily affected as well. The direct and indirect effects of healthcare from these factors is left as an exercise to the reader.

2 Thailand's Healthcare System

Thailand's healthcare system is overall considered one of the best globally, with a rank of 5 out of 195 countries according to **ThailandGHSIndex**. This composite ranking considers a wholistic perspective on the healthcare system as a whole, including elements such as the accessibility of healthcare, the availability of prehospital care, disease response capability and detection capacity among other metrics. However, these metrics paint a different story than the picture that most Thais (and HCWs) see of the healthcare system



Figure 1: The typical patient load in public hospitals hospice

on a daily basis, with overcrowding, long wait times and, in distant districts, a distinct lack of HCWs.

2.1 Inequity of Healthcare

The availability of healthcare within countries is not distributed evenly. The highest proportion of healthcare workers (HCWs) tend to be found in the most developed areas. This is especially true for Thailand, where the “primate city” of Bangkok serves as a hub for a large number of world-class healthcare facilities, while remote provinces often receive care that may be considered as below the standard set by big cities. In fact, over four million people in Thailand alone are without ready access to healthcare services **tilakulThailandsHealthSystem2026**. Factors driving this inequity to healthcare include the lack of homogeneity in development, the selection pool for training of HCWs, uneven distribution of funding and human factors among others. While initiatives such as those by the NHSO exist to allow patients access to hospital, they are often found to have limited scope in practice, resulting in ineffective solutions to the problem of healthcare access. This is shown best in **medrole**, which shows how doctors are distributed unevenly to the patient population, and that workload differs greatly across healthcare districts. One potential solution involves increasing funding and availability of healthcare centers in distant provinces, however, the feasibility of such solutions often depend on individual community context. Consequently, a comprehensive solution remains notably absent.

2.2 Pressing Challenges

3 Social Context

Healthcare is more than just treating symptoms and prescribing medicines for illness. Effective healthcare demands the involvement of HCWs in the com-

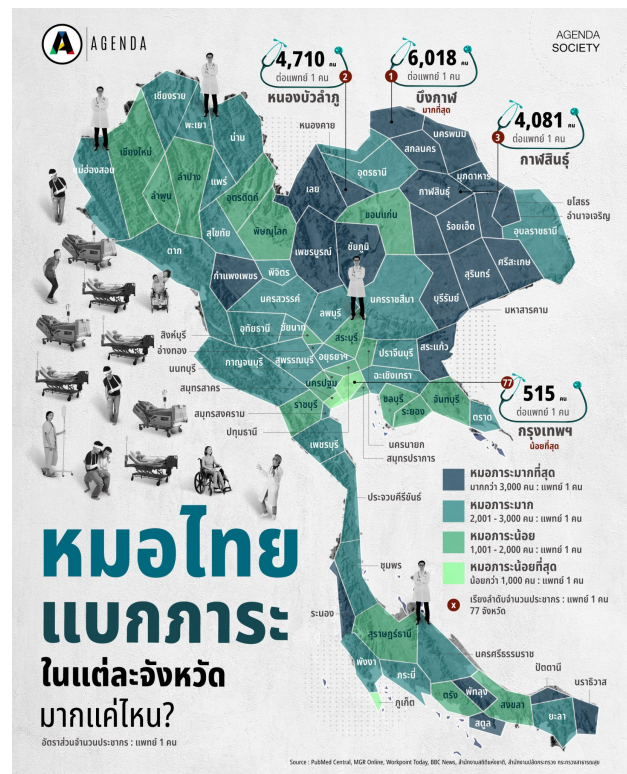


Figure 2: Distribution of healthcare availability across healthcare districts in Thailand medrole

munity, both at the individual and community scale. Consequently, community relationships serve as a key method for the prevention of disease and to increase the availability of healthcare within communities, especially those in remote areas. Thailand is unique in this context, with a large network of community HCWs known as “village health volunteers (VHVs)” who serve to protect the health of their communities, monitor vulnerable residents and effectively report community health status to the relevant authorities to promote informed policy decisions that reflect community status at both the community and country scale by narrowing the gap between policymakers and residents of the local area. (One way to define the decentralization of healthcare is the term “community based care”)

Community based care is an effective system, though it has some limitations, including the risk of decreasing equitability of care, to the limited technology available for VHVs. Solutions to these problems are few and far between, but could hold the potential to be highly effective and beneficial for the community at large.

3.1 Expectations of Healthcare

3.2 The Mental Health Stigma

3.3 Healthy Habits

One challenge to providing effective preventative healthcare for the general population is the lack of action regarding healthy eating habits. The typical

Thai meal does not cover all nutritional bases, and often exceeds healthy levels for many macronutrients, including calories, carbohydrate, fats, salt and sugar. The habit of choosing food that is highly flavorful and relatively in excess compared to the nutritional requirements may result in long-term consequences such as the increased risk for NCDs, including renal, cardiovascular and diabetes related complications.

4 Challenges for the Elderly

4.1 Non-communicable Diseases

4.2 Rehabilitation Medicine

4.3 Life Quality & Interventions

The quality of life of a population is determined by a wide range of factors, including social environment, region, social equitability and personal factors among others. Though over 30 years has passed, the Dahlgren and Whitehead remains one of the standard ways of understanding the influences of health even before a visit to the medical practitioner. Many solutions for wellbeing do not start in the hospital, but instead in the community, where interventions are often more effective than treatment of already developed conditions. The prevention of disease, especially non-communicable diseases remains more cost effective than treatment. However, many limitations remain that result in social interventions being overlooked. Solutions in this field are direly needed, and may serve to actively enable the prevention of diseases in the subclinical phase, resulting in healthier communities and longer healthspans for the population.

Any effective interventions on life quality for the population, especially in the elderly population, must take into account the feasibility of these interventions in actual practice. Many a lesson has been learned by well-meaning NGOs and even governmental aid organizations that intervention effectiveness is not determined by the quality of each product or commercial, but the feasibility of adoption to daily life is the factor which indicates if interventions will prove successful in the long run. One example of this is the well-meaning diabetes diet and exercise regimens, which often face failure as a consequence of how patients are not able to adhere to them, and often end up reverting to their original, disease causing, diets. Therefore, one could speculate that more realistic interventions based on individual ability could drive better compliance and greater success **samiEffectDietType2017**. Put simply, behavioral friction kills effective interventions and technology. For further details on policy interventions, please see **weaverWillItWork**.

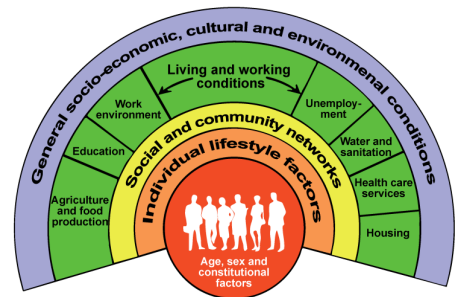


Figure 3: Model of the social determinants of health (Dahlgren and Whitehead, 1991) from Mariette-JB, n.d.

Table 1: Example table

Name		
First Name	Last Name	Grade
John	Doe	7.5
Richard	Miles	5

5 Emerging Technologies

Overall, artificial intelligence (AI) is emerging as a powerful tool for change in healthcare systems and technology. From AI text transformers to advanced data processing and analytics, AI holds the potential to improve access to healthcare and increase the efficiency of Thailand’s medical infrastructure. However, several barriers still hinder the widespread adoption of AI in healthcare. These include technological complexity, lack of cultural sensitivity, and a failure to address the specific needs of the Thai elderly population and their families.

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- Second item in a list
- Third item in a list

First This is the first item

Last This is the last item

5.1 Subsection

1. First numbered item in a list
2. Second numbered item in a list
3. Third numbered item in a list

Bibliography

Mariette-JB (n.d.). *The Health and Wellbeing of People in Europe*. Infra4NextGen. URL: <https://infra4nextgen.com/resources/data-summaries/healthy/2-theoretical-model/> (visited on 05/17/2026).